

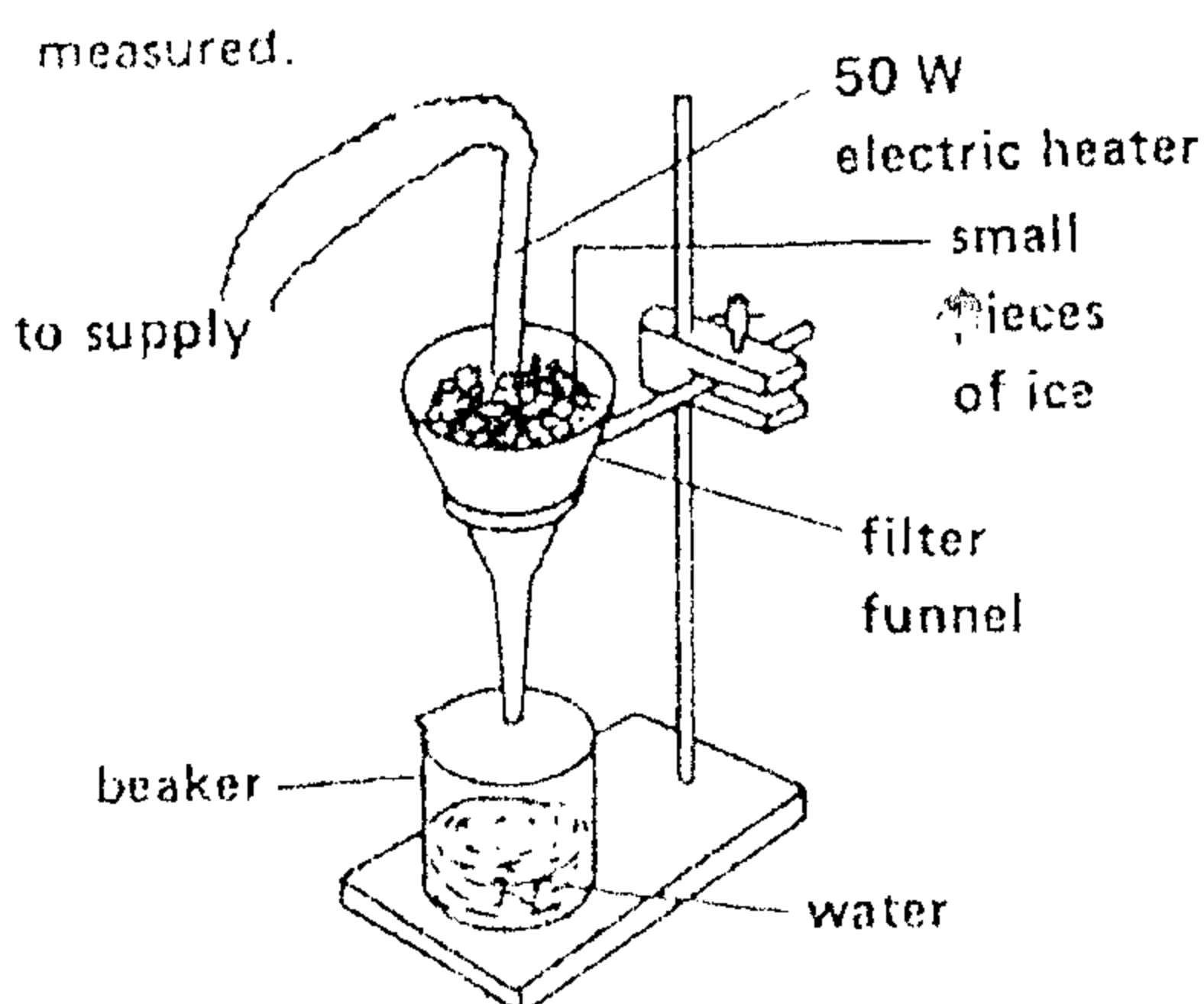
1. The unit of electrical resistance may be expressed as

(A) $1\Omega = 1\text{ V}\cdot\text{A}^{-1}$
 (B) $1\Omega = 1\text{ A}\cdot\text{V}$
 (C) $1\Omega = 1\text{ A}\cdot\text{V}^{-1}$
 (D) $1\Omega = 1\text{ W}\cdot\text{A}^{-1}$

2. 1 gram is equal to

(A) 10 milligrams
 (B) 100 milligrams
 (C) 1 000 milligrams
 (D) 10 000 milligrams

3. By switching on the heater in the following apparatus for a known time and weighing the water collected in the beaker, the specific latent heat of fusion of ice can be measured.



Why is this method very inaccurate when used in the Caribbean?

- (A) The heater is not powerful enough to melt the ice.
 (B) The ice would melt even if the heater were not turned on.
 (C) The temperature of the ice is 30°C instead of 0°C .
 (D) Water would be evaporating from the open beaker.

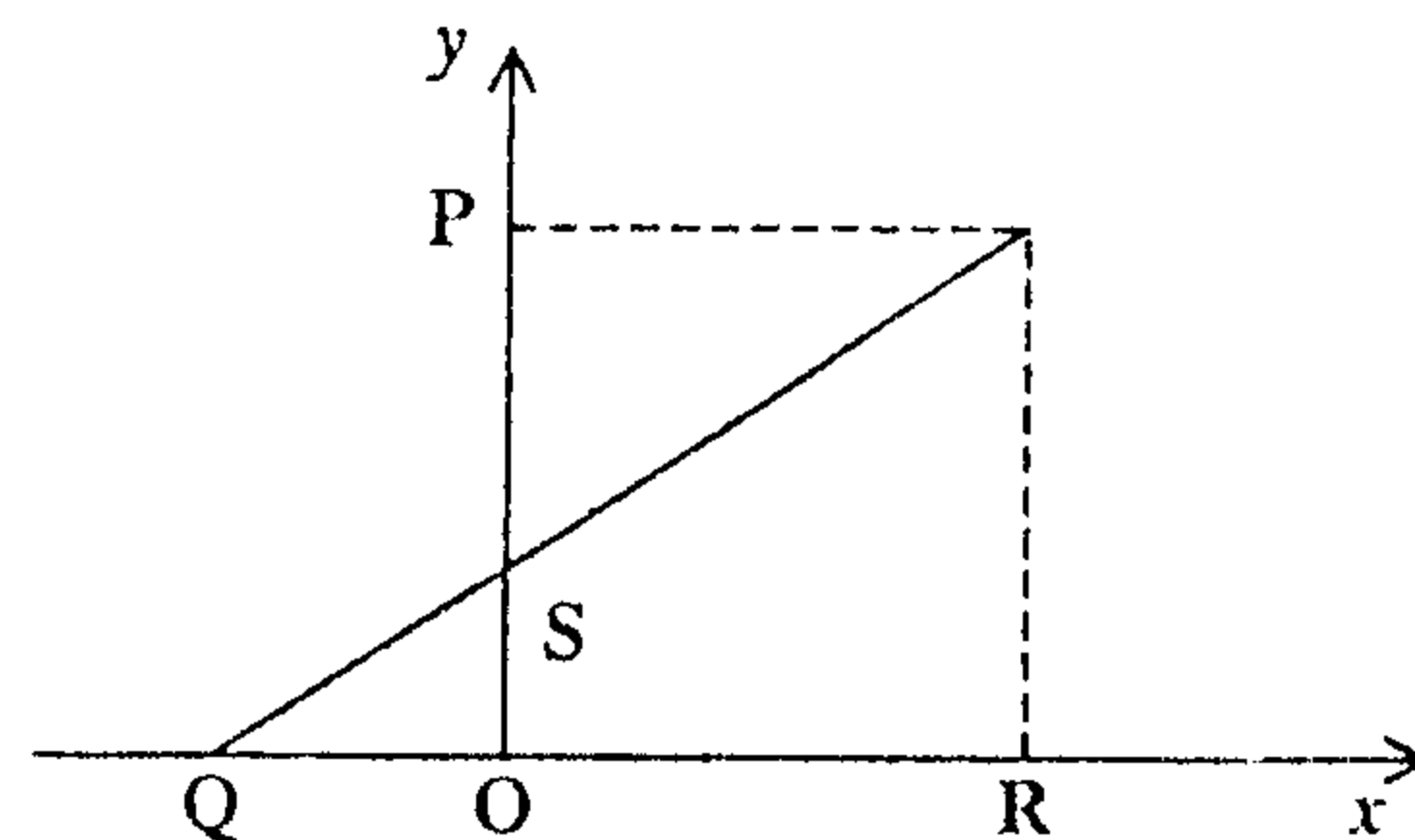
4. Which of the following units is the unit of momentum?

(A) kg s^{-1}
 (B) kg m s^{-1}
 (C) kg m s^{-2}
 (D) N m

5. Which of the following quantities is dimensionless?

(A) Density
 (B) Mass
 (C) Relative Density
 (D) Volume

Item 6 refers to the diagram below.



6. For any point (x, y) on the line QS, when $x = 0$, the value of y is

(A) OQ
 (B) OS
 (C) OP
 (D) OP/OR

7. An object of mass, m , is attached to a spring balance and its weight, w , recorded. What will be the result if the object is taken to the moon and weighed there?

(A) Mass = m ; weight greater than w
 (B) Mass = m ; weight less than w
 (C) Mass greater than m ; weight = w
 (D) Mass less than m ; weight = w

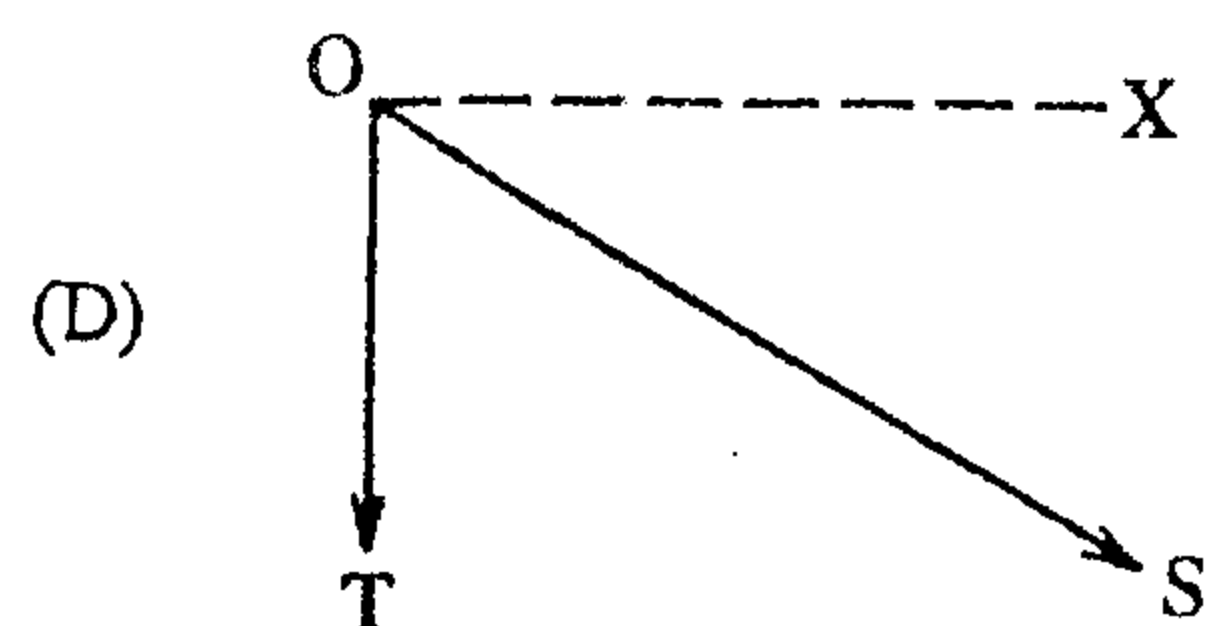
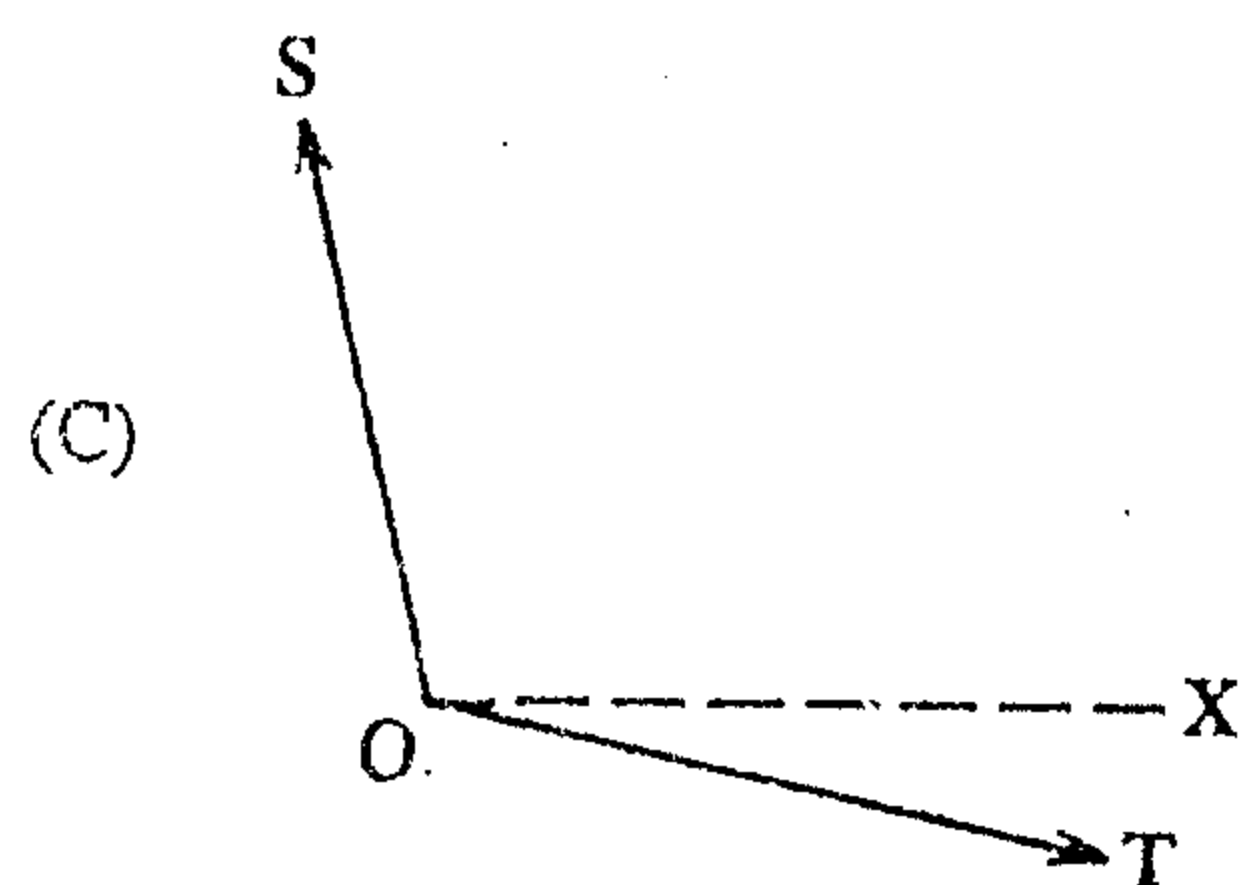
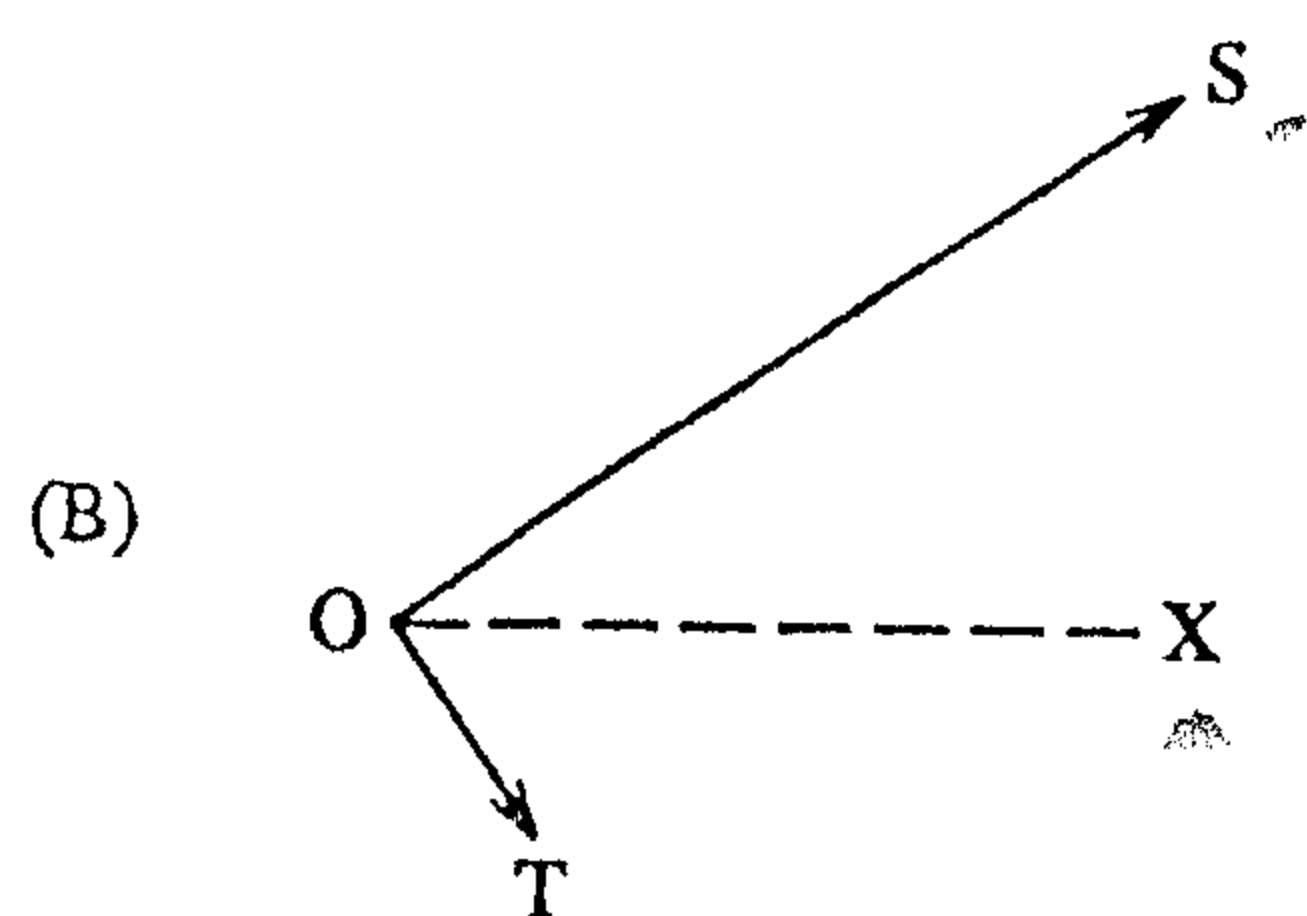
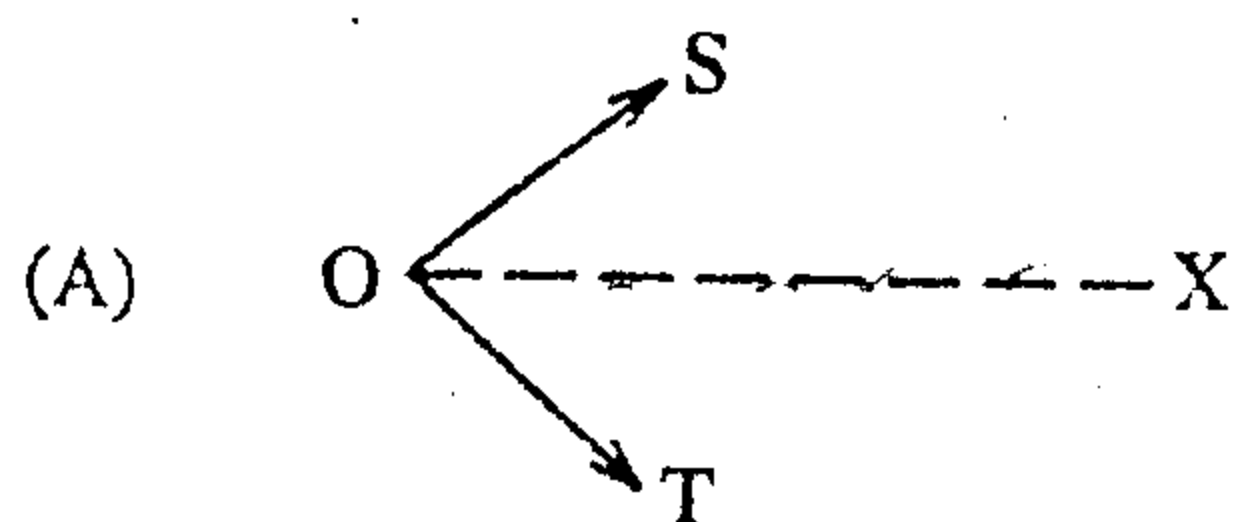
8. Which of the following quantities will be constant, if a constant net force is applied to a body?

(A) Velocity
 (B) Kinetic energy
 (C) Momentum
 (D) Acceleration

9. A stable well designed racing car must have a

(A) low centre of gravity
(B) narrow wheel base
(C) sun roof
(D) long front

10. The diagrams below, drawn to scale, represent two forces S and T acting at O. In which of the following diagrams is the resultant in the direction, OX?



11. Which of the following formulae could be used to find the average speed of an object?

(A) $\frac{\text{Change in velocity}}{\text{Time taken}}$
(B) $\frac{\text{Change in displacement}}{\text{Time taken}}$
(C) $\frac{\text{Distance travelled}}{\text{Time taken}}$
(D) Distance travelled x Time taken

12. Which of the following measuring devices is suitable for measuring the diameter of a human hair?

(A) Tape measure
(B) Vernier calipers
(C) Micrometer screw gauge
(D) Metre rule

13. When two bodies collide, momentum is conserved. This means that the

(A) kinetic energy before impact is equal to that after impact
(B) momentum of each body is unchanged
(C) algebraic sum of the velocities before impact is equal to the sum of the velocities after impact
(D) total momentum of the bodies before impact is equal to the total momentum of the bodies after impact

14. Power can be defined as

(A) force x distance moved
(B) $\frac{\text{force}}{\text{time}}$
(C) $\frac{\text{work done}}{\text{time}}$
(D) work done x time

15. An object is removed from the ground and placed on a shelf. Which of its properties would you expect to change?

(A) Mass
(B) Volume
(C) Potential energy
(D) Kinetic energy

16. Pressure, P , in a liquid can be calculated from the formula

$$P = \rho gh$$

Which sets of units below will give the pressure in the SI unit?

	ρ	g	h
(A)	g cm^{-3}	m s^{-2}	mm
(B)	kg m^{-3}	N kg^{-1}	m
(C)	g cm^{-3}	N kg^{-1}	m
(D)	kg m^{-3}	cm s^{-2}	cm

17. A bubble of gas rises to the surface of a soft drink. This is because the

(A) upthrust on the bubble is greater than the weight of the bubble
(B) upthrust on the bubble is greater than the weight of water it displaces
(C) weight of water displaced by the bubble is less than the weight of the bubble
(D) density of the gas is greater than the density of the drink

18. A glass bulb is filled with a gas at a temperature of 293 K. If the initial pressure of the gas is P , what will it be when the temperature increases to 360 K?

(A) $\frac{293}{360} \times P$
(B) $\frac{360}{293} \times P$
(C) $\frac{293}{360} \times \frac{1}{P}$
(D) $\frac{360}{293} \times \frac{1}{P}$

19. Boyle's law for a gas can be tested experimentally, provided which of the following is/are maintained constant?

I. Temperature
II. Pressure
III. Density
IV. Mass

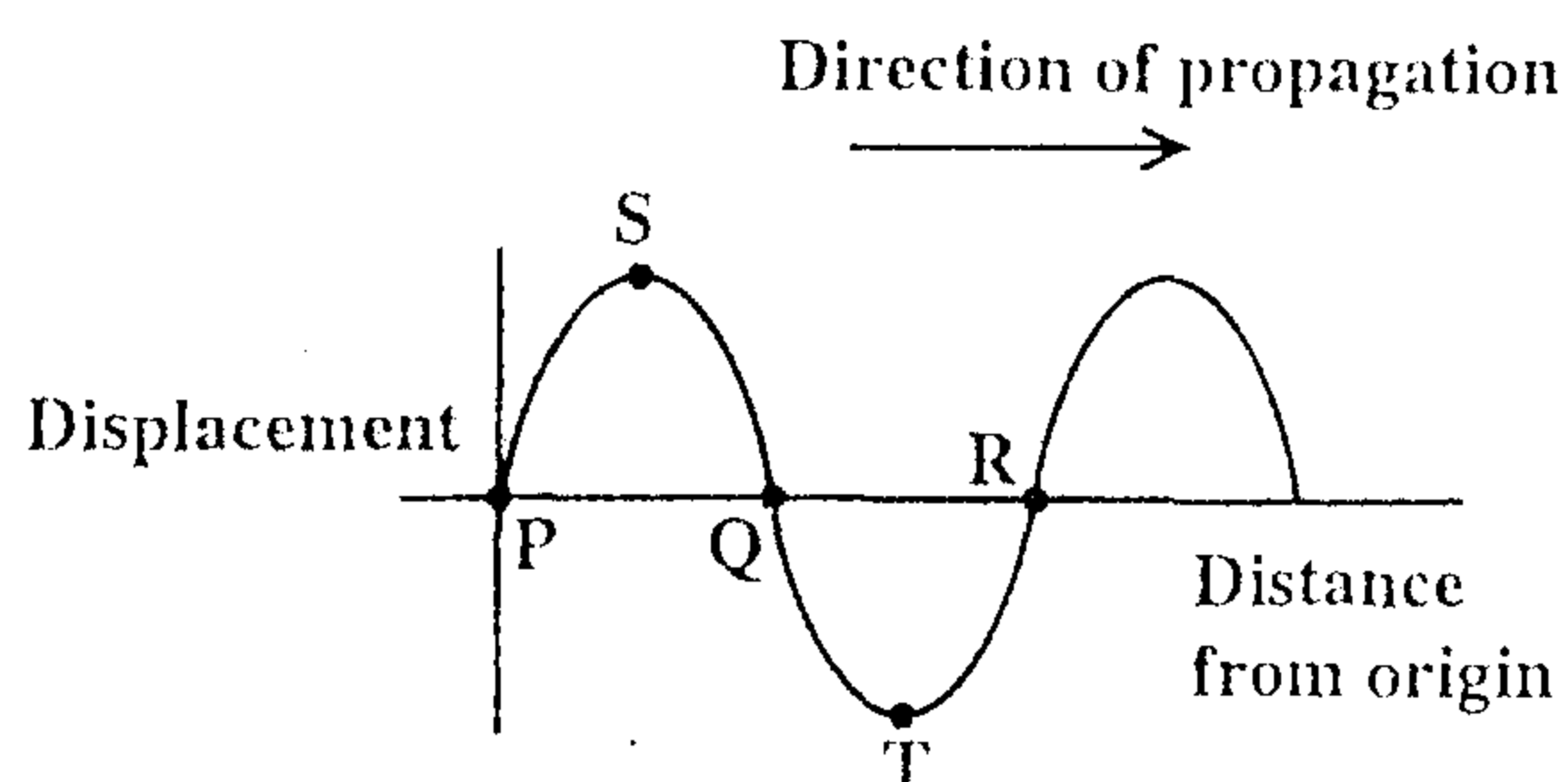
(A) III only
(B) I and II only
(C) I and IV only
(D) I, II and III only

20. Which of the following is the MOST suitable range for a clinical thermometer?

(A) 0°C to 44°C
(B) -10°C to 110°C
(C) 35°C to 100°C
(D) 35°C to 44°C

21. The specific heat capacity of a material is the energy required to
- (A) melt 1 kg of the material with no change of temperature
 - (B) change the temperature of the material by 1 K
 - (C) change 1 kg of the liquid material to 1 kg of gas without a change in temperature
 - (D) change the temperature of 1 kg of the material by 1 K
22. The specific latent heat of vaporization of water is the energy required to change 1 kg of water at
- (A) 0 °C to steam at 100 °C
 - (B) 99.9 °C to steam at 100.1 °C
 - (C) 100 °C to steam at 100 °C
 - (D) 0 °C to ice at 0 °C
23. Which of the following statements concerning the radiation of heat is true?
- I. Radiation can only take place in a material medium.
 - II. A good absorber is also a good emitter of radiation.
 - III. Dark dull surfaces are better emitters than shiny ones.
- (A) III only
 - (B) I and II only
 - (C) I and III only
 - (D) II and III only
24. An electric kettle full of water is plugged into the mains. The process by which heat travels through the water is
- (A) electrification
 - (B) evaporation
 - (C) convection
 - (D) radiation
25. There are NO attractive forces between the molecules in a
- (A) liquid and gas
 - (B) solid and a liquid
 - (C) liquid
 - (D) gas
26. In a Brownian motion experiment with a smoke cell, the fast random motion of the smoke particles which is observed is due to the
- (A) collision of smoke particles with air molecules in the cell
 - (B) bombardment of one smoke particle by other smoke particles
 - (C) regular vibration of the smoke particles
 - (D) collision of the smoke particles with the walls of the container
27. Which of the following is the correct relation between the wavelength λ , speed v , and frequency f , of a wave?
- (A) $\lambda = f v$
 - (B) $f = \frac{\lambda}{v}$
 - (C) $f = \frac{v}{\lambda}$
 - (D) $\lambda = \frac{f}{v}$
28. The spreading of a wave into the region behind an obstruction is called
- (A) diffraction
 - (B) refraction
 - (C) superposition
 - (D) interference

29.



Which of the following statements about the wave shown in the diagram above is/are true:

- I. Points P, Q and R are in phase.
- II. Points S and T are out of phase.
- III. The wavelength of the wave is the distance PR.

- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only

30. If sounds of differing frequencies are played on a piano, in which of the following characteristics would a change be detected?

- (A) Loudness
- (B) Speed
- (C) Pitch
- (D) Wavelength

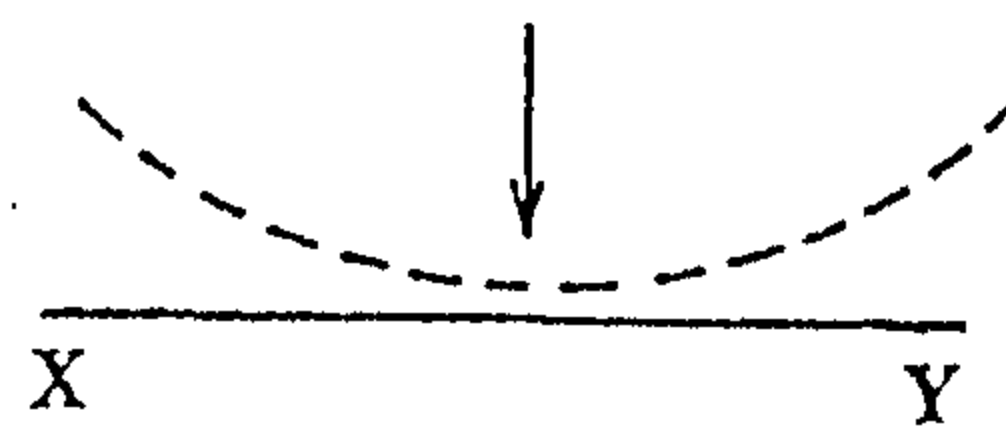
31. An explosion causes the emission of the following types of radiation.

- I. Light
- II. Sound
- III. Infra-red

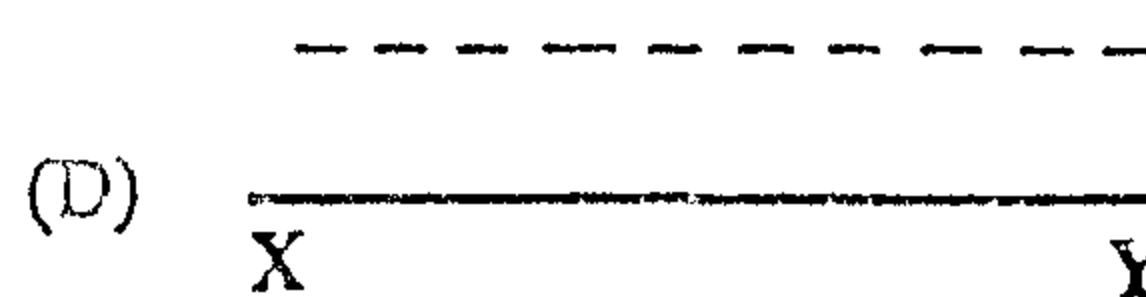
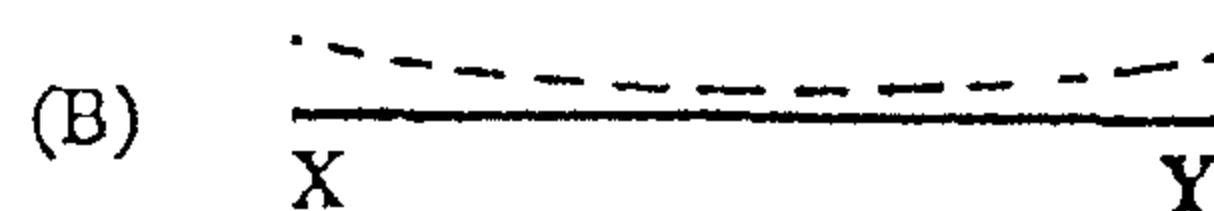
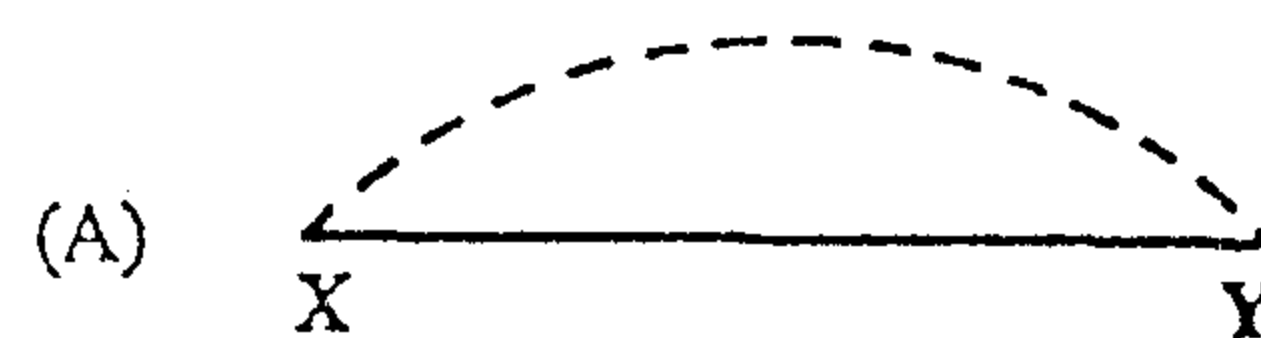
Which of these will be received first by a person some distance away?

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

32.



The figure above shows a wavefront about to reach a reflecting surface XY. Which of the following would represent the wave-front after reflection?



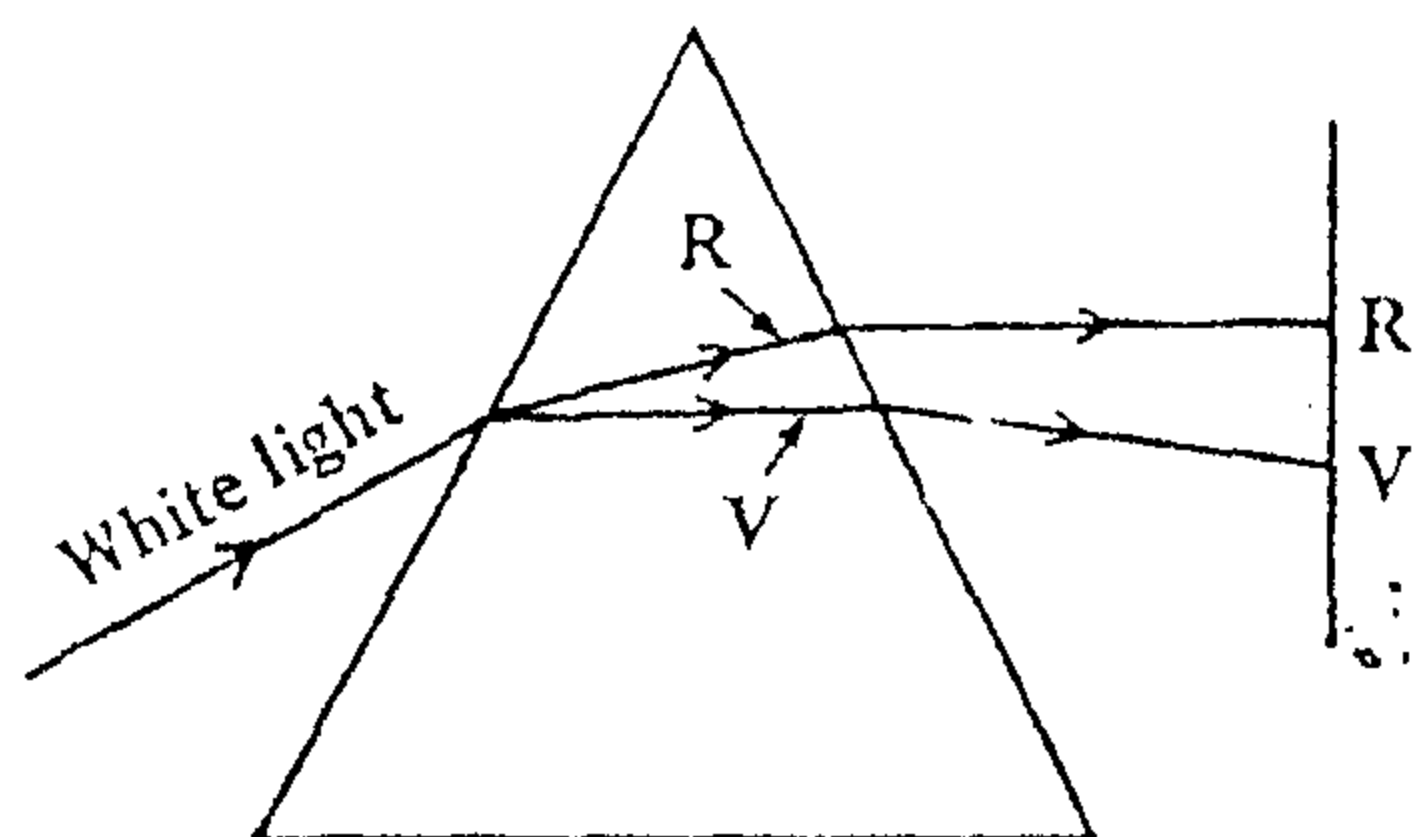
33. The sharp edges of shadows suggest that light

- (A) has a wave nature
- (B) is a form of energy
- (C) travels in straight lines
- (D) travels very quickly

34. The specific latent heat of fusion of water is 340 kJ kg^{-1} . This means that when 10 kg of water freezes

- (A) 34 kJ of heat is absorbed
- (B) 34 kJ of heat is given out
- (C) 3 400 kJ of heat is absorbed
- (D) 3 400 kJ of heat is given out

35.



The diagram above shows a ray of white light being dispersed by a prism to form a visible spectrum. Which of the following makes this possible.

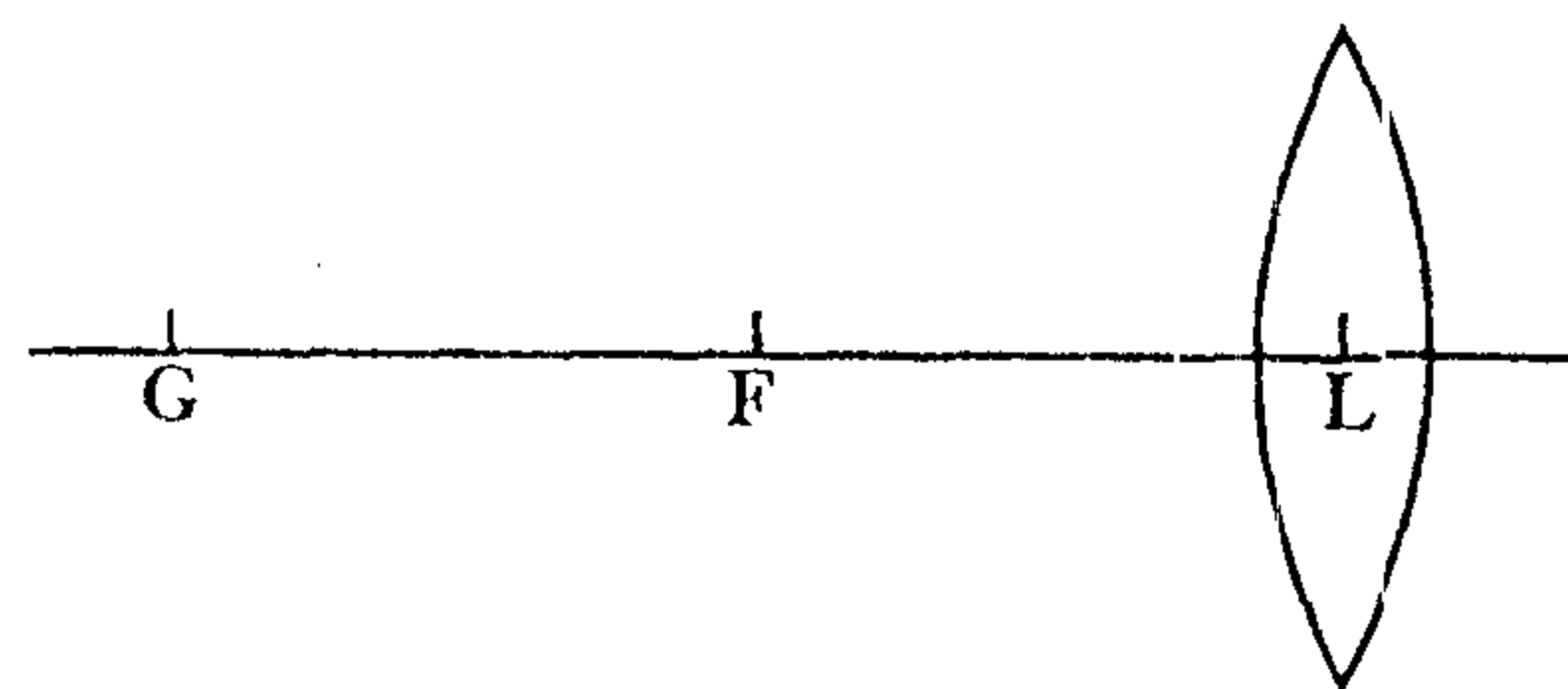
- (A) The colour violet has the shorter wavelength, hence refracts more than colour red.
- (B) The colour red has the longer wavelength, hence refracts more than colour violet.
- (C) The colour violet has the longer wavelength, hence refracts more than colour red.
- (D) The colour red has the shorter wavelength, hence refracts more than colour violet.

36. Which of the following statements would be true of the image of an object placed at the bottom of a tank of water and viewed vertically from above?

- I. It is virtual.
- II. It is diminished.
- III. It is nearer to the eye than the object.

- (A) IV only
- (B) I and II only
- (C) I and III only
- (D) II and III only

37.



The above diagram represents a thin converging lens, L, with a principal axis GL. FL is the focal length of the lens and $GF = FL$. Where should an object be placed so as to produce a virtual magnified image?

- (A) At G
- (B) To the left of G
- (C) Between F and G
- (D) Between L and F

38. Which of the following would be true of the image formed by the lens of a camera?

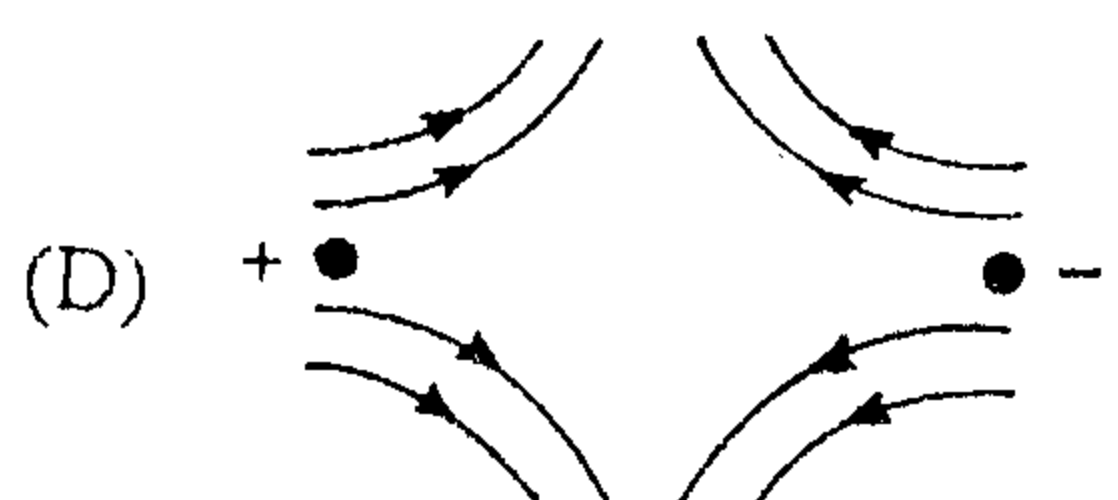
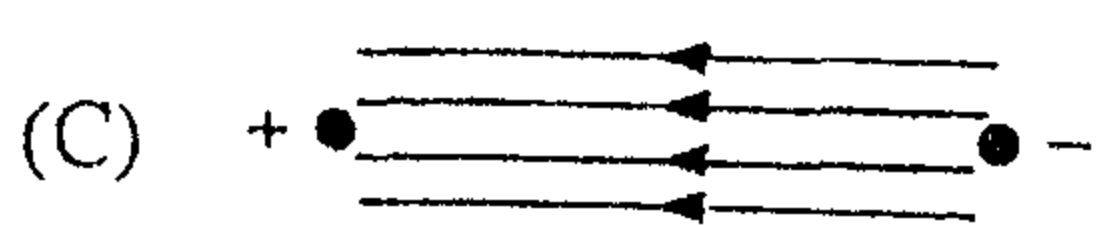
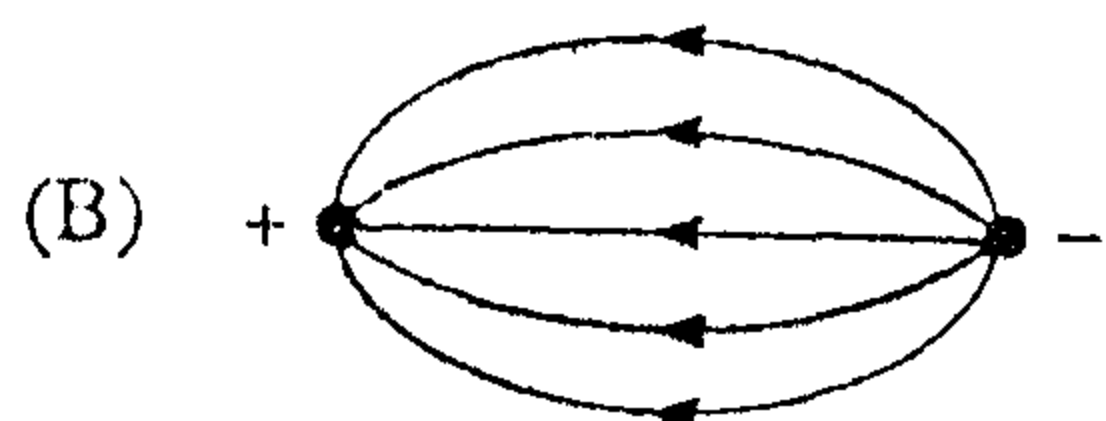
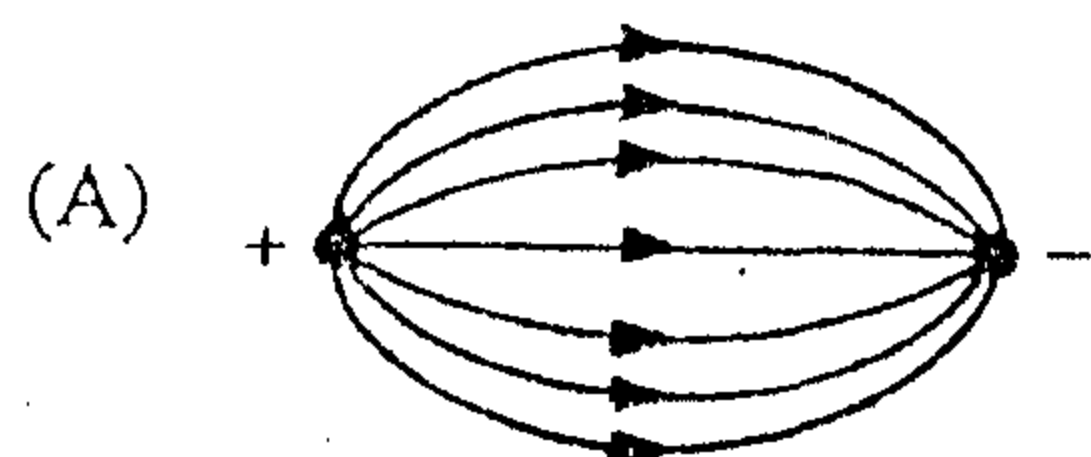
- I. It is virtual.
- II. It is real.
- III. It is diminished.
- IV. It is upright

- (A) I only
- (B) II and III only
- (C) I and IV only
- (D) IV only

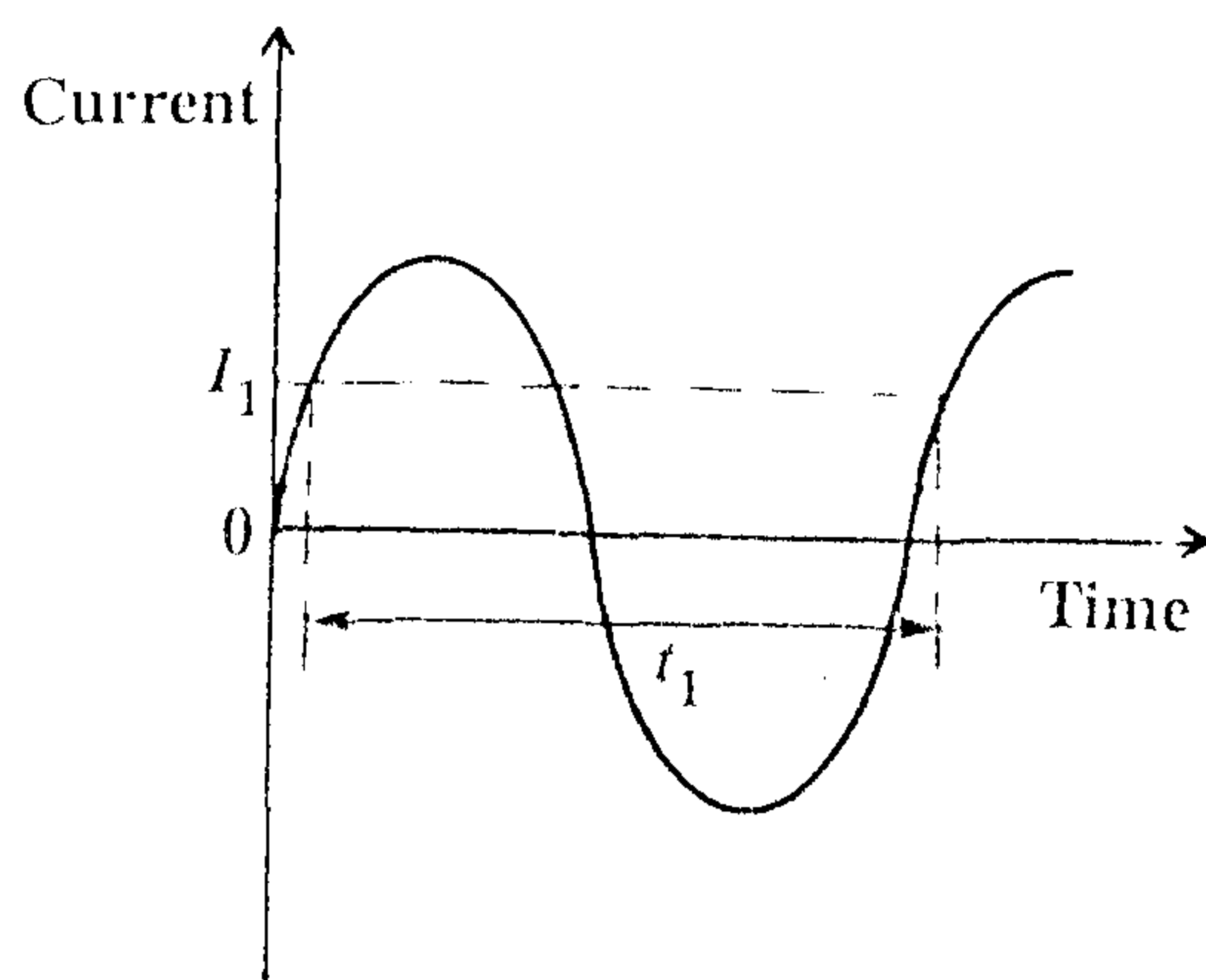
39. A glass rod is rubbed with a piece of silk and becomes positively charged. The glass rod became charged by

- (A) gaining protons
- (B) losing protons
- (C) gaining electrons
- (D) losing electrons

40. Which of the following diagrams represents the electric field existing between two oppositely charged point charges?



41.



The graph above shows the variation of alternating current with time. The value frequency of this current is

- (A) t_1
 (B) $1/t_1$
 (C) I_1
 (D) $1/I_1$

42. Which of the following could NOT be a unit of current?

- (A) W V^{-1}
 (B) $\text{V } \Omega^{-1}$
 (C) C s^{-1}
 (D) C s

43. Which of the following equations CANNOT be used to determine the power dissipated in a resistor?

- (A) $P = I^2 R$
 (B) $P = \frac{V}{R^2}$
 (C) $P = VI$
 (D) $P = \frac{R}{V^2}$

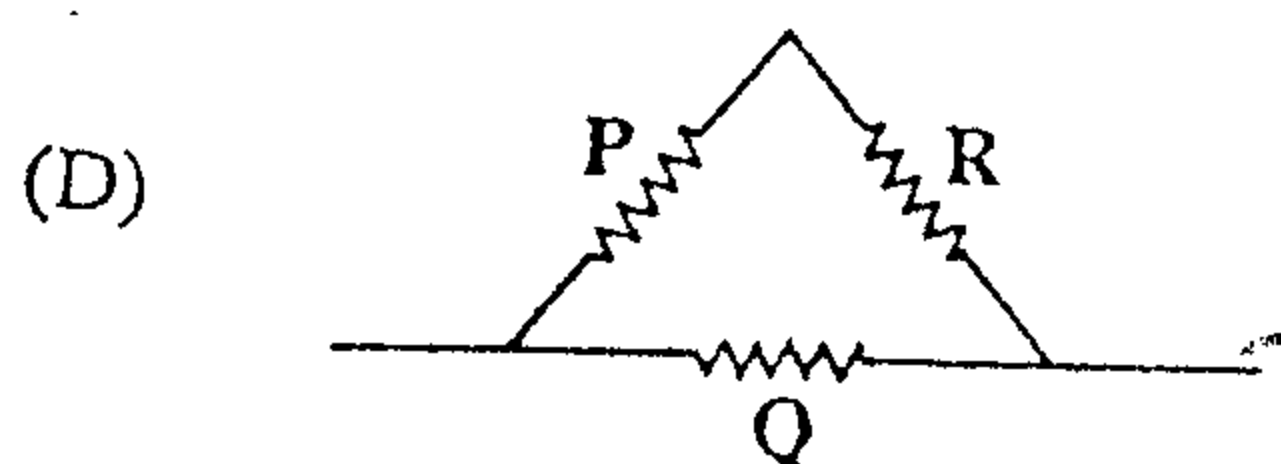
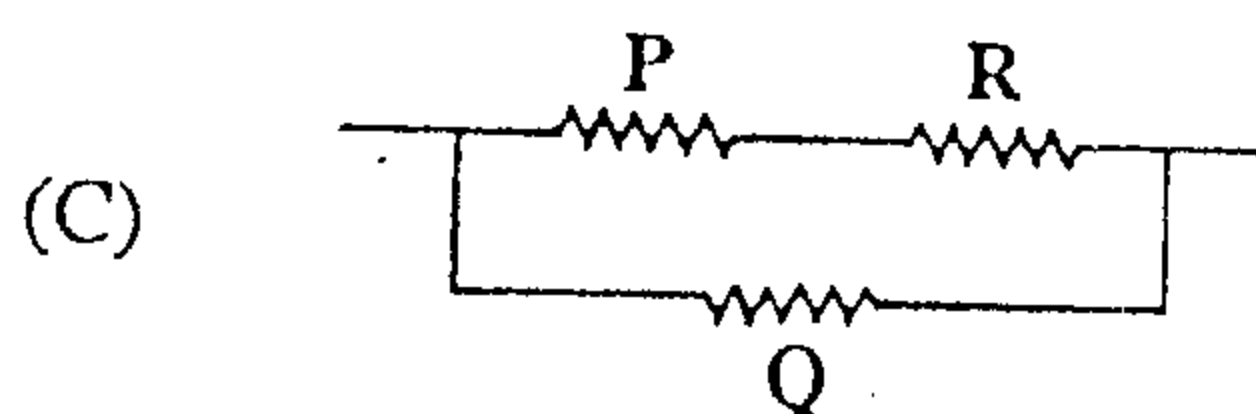
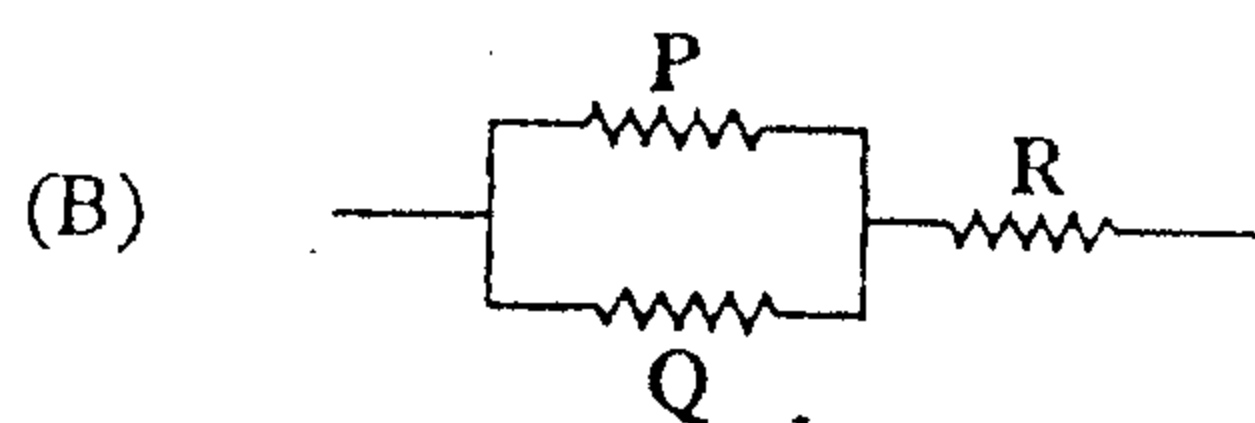
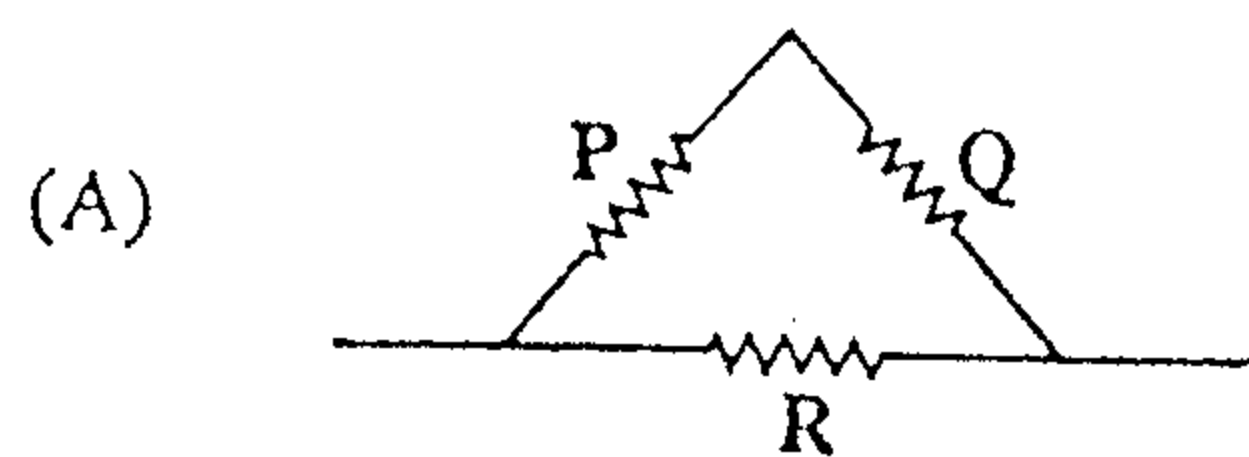
44. The refractive index of a transparent medium with a critical angle, c , for light travelling from the medium to air is

- (A) $\frac{1}{c}$
 (B) $\frac{90^\circ}{\sin c}$
 (C) $\frac{\sin 90^\circ}{\sin c}$
 (D) $\sin c$

45. In domestic installation systems, which of the following should be placed in the live wire?

- I. Switches
 II. Circuit breakers
 III. Fuses
- (A) I only
 (B) III only
 (C) II and III only
 (D) I, II and III

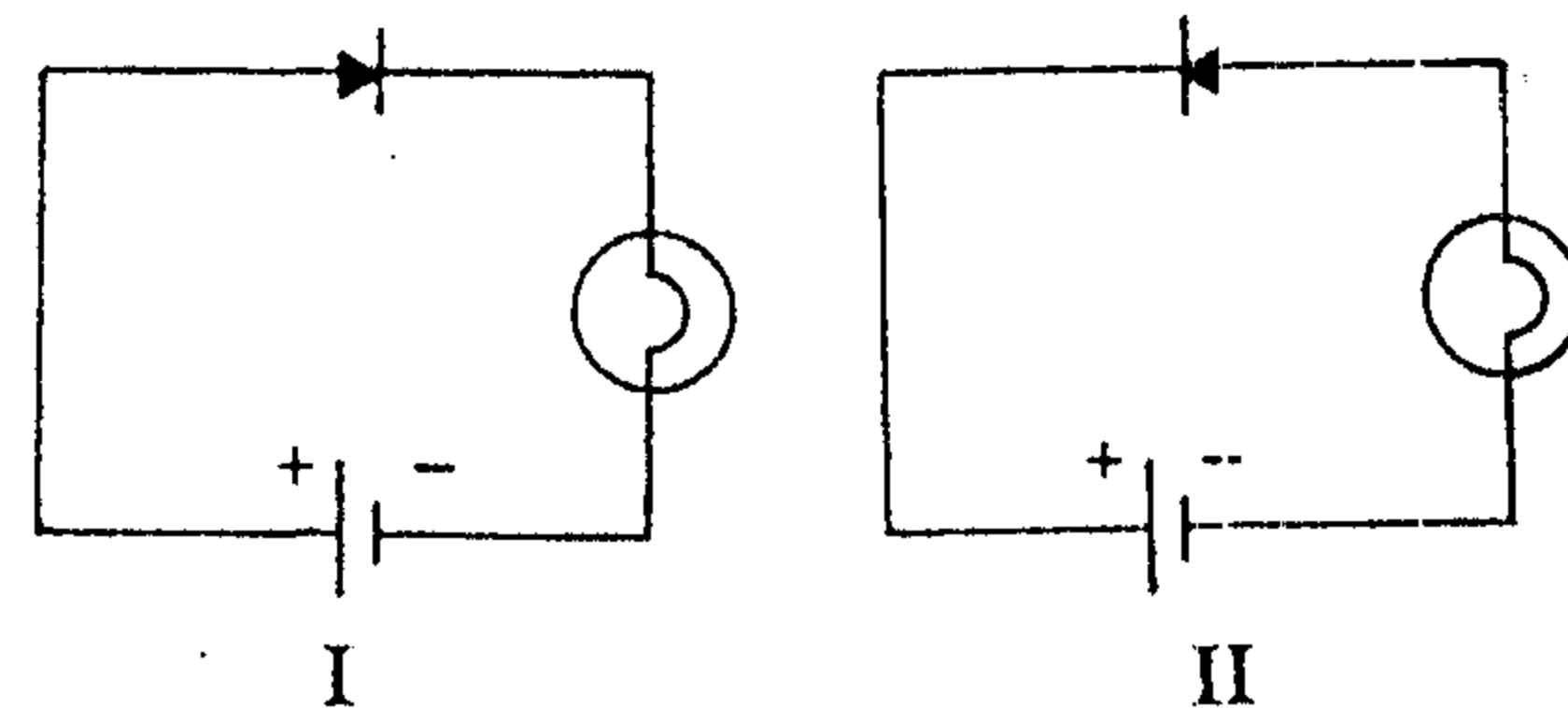
46. In which of the following diagrams are resistors P and Q in series with each other and parallel with R?



47. When a large current passes through a fuse, which of the following is the sequence of events?

- (A) Wire gets hot $\rightarrow$ wire melts $\rightarrow$ current is cut off
 (B) Wire gets hot $\rightarrow$ current is cut off $\rightarrow$ wire melts
 (C) Wire melts $\rightarrow$ current is cut off $\rightarrow$ wire gets hot
 (D) Wire melts $\rightarrow$ wire gets hot $\rightarrow$ current is cut off

48.



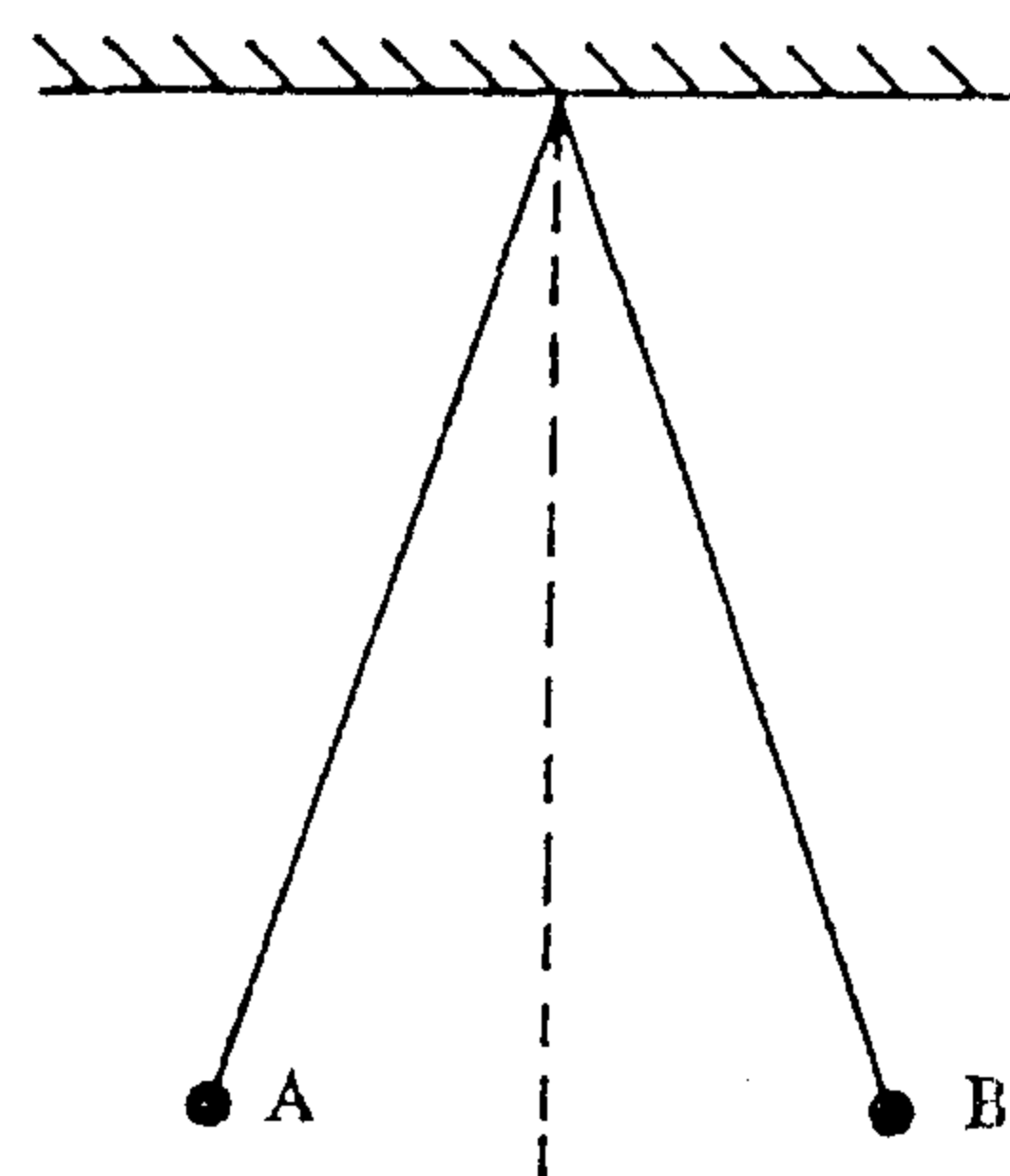
A simple experiment was conducted using the circuit diagrams shown above. The same components were used and the bulb was lit to normal brightness in each case.

Which of the following statements would be correct?

- I. The bulb is defective.
 II. The battery is defective.
 III. The diode is defective.

- (A) I only
 (B) III only
 (C) I and II only
 (D) II and III only

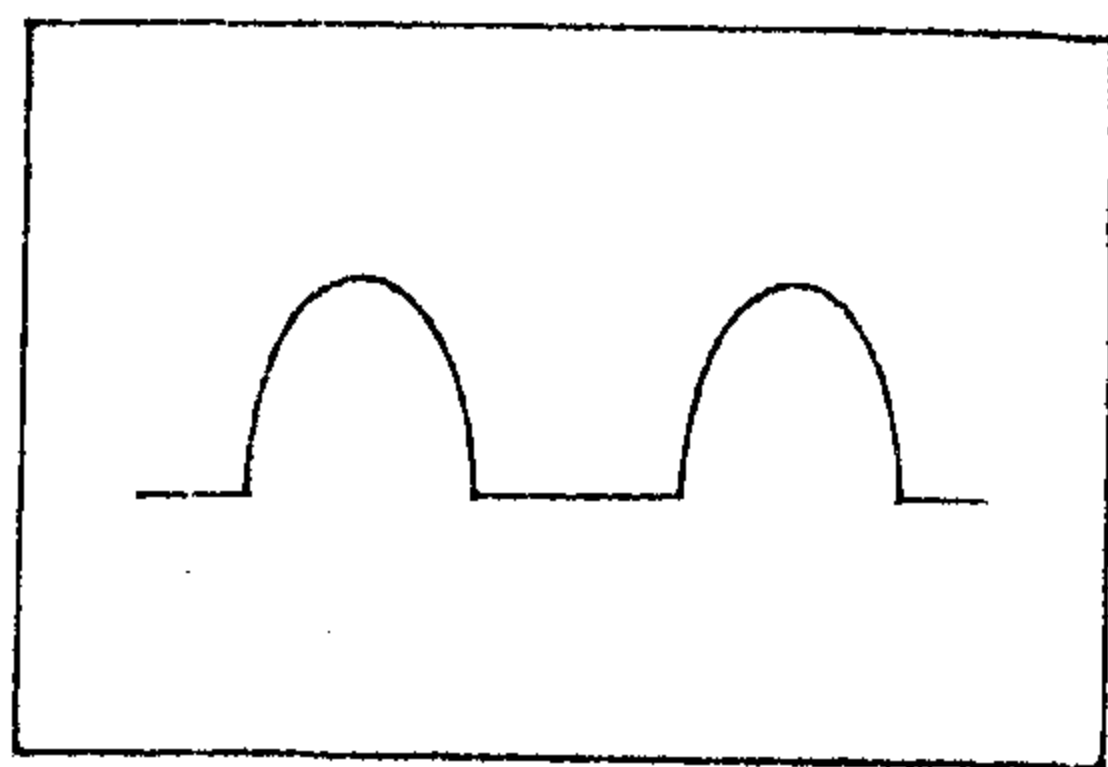
49.



Two light, aluminium spheres A and B are suspended by insulating threads. If they come to rest as shown in the diagram, the force keeping them apart is

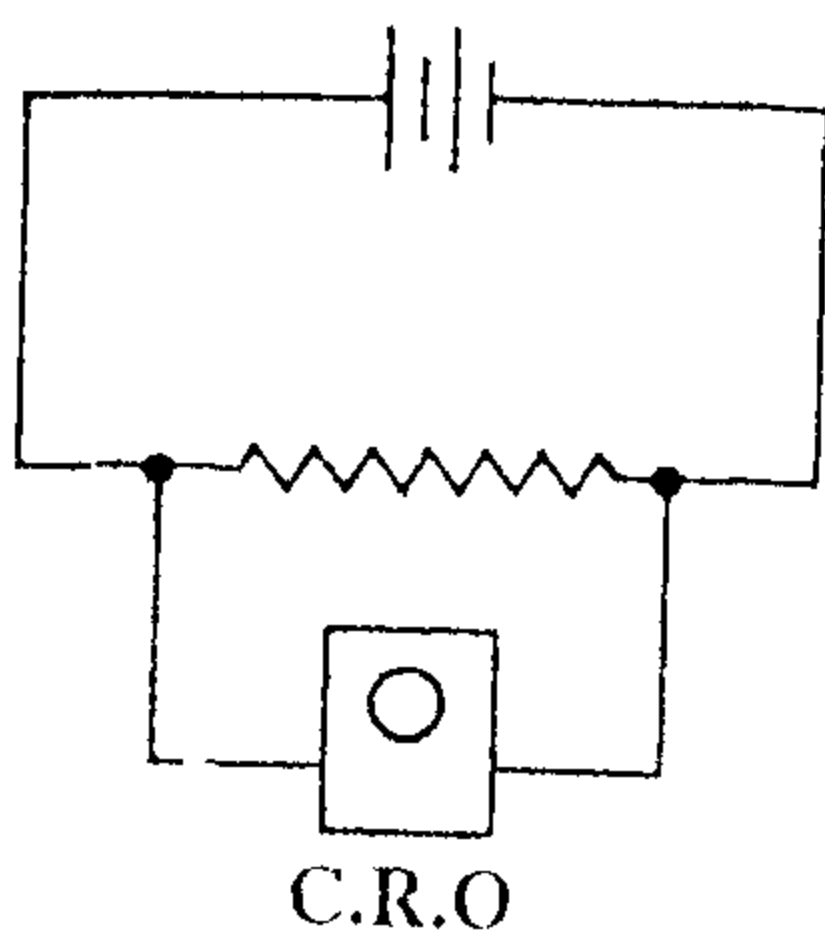
- (A) gravitational
 (B) electrostatic
 (C) magnetic
 (D) centripetal

50.

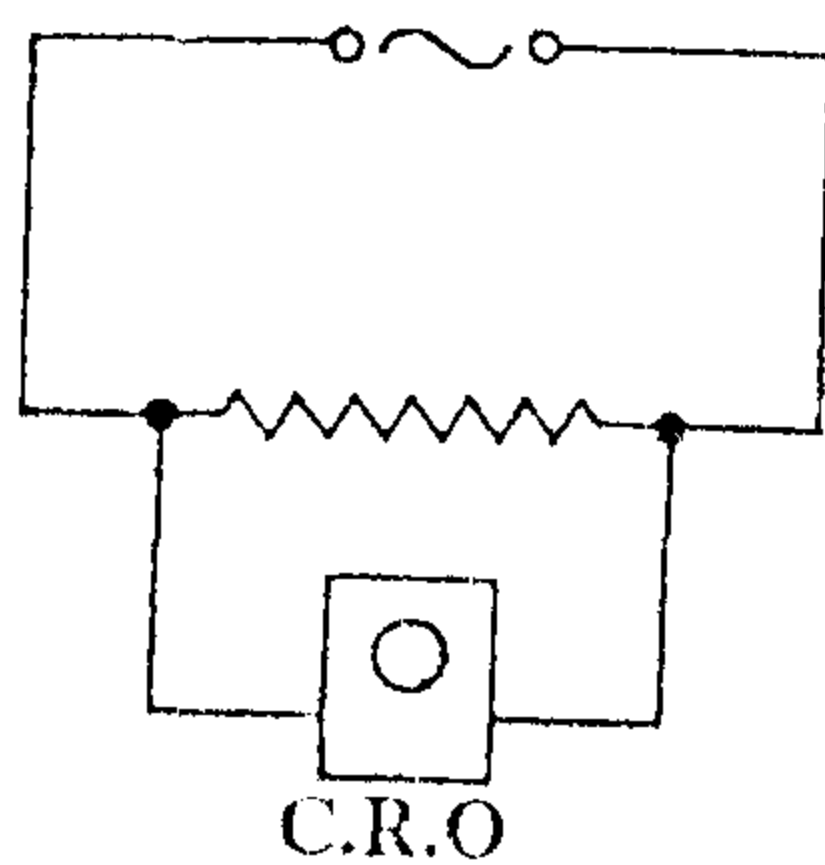


Which of the circuits below could produce the trace on the oscilloscope screen shown in the diagram above when the ends of the resistor are connected to the Y plates?

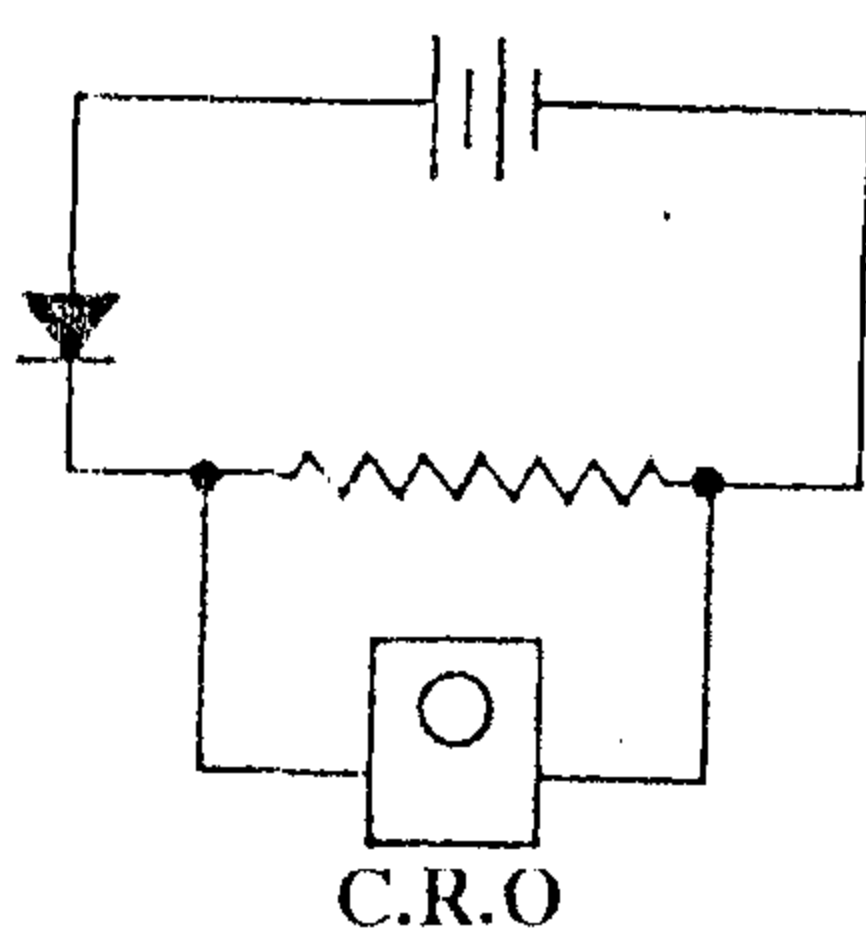
(A)



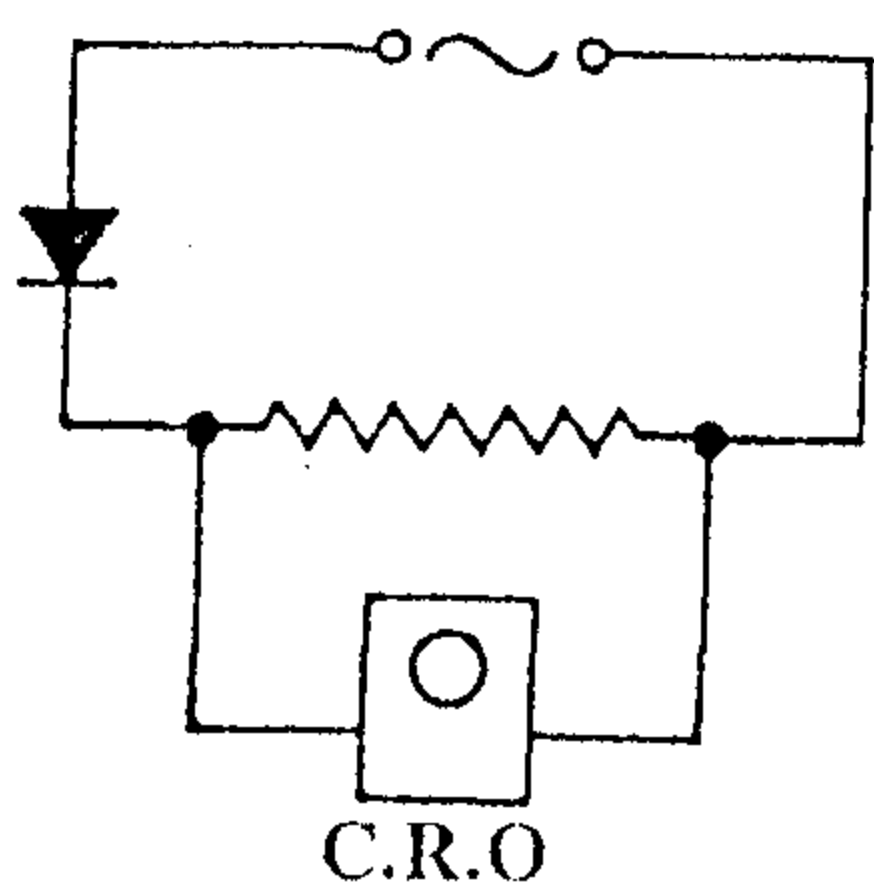
(B)



(C)

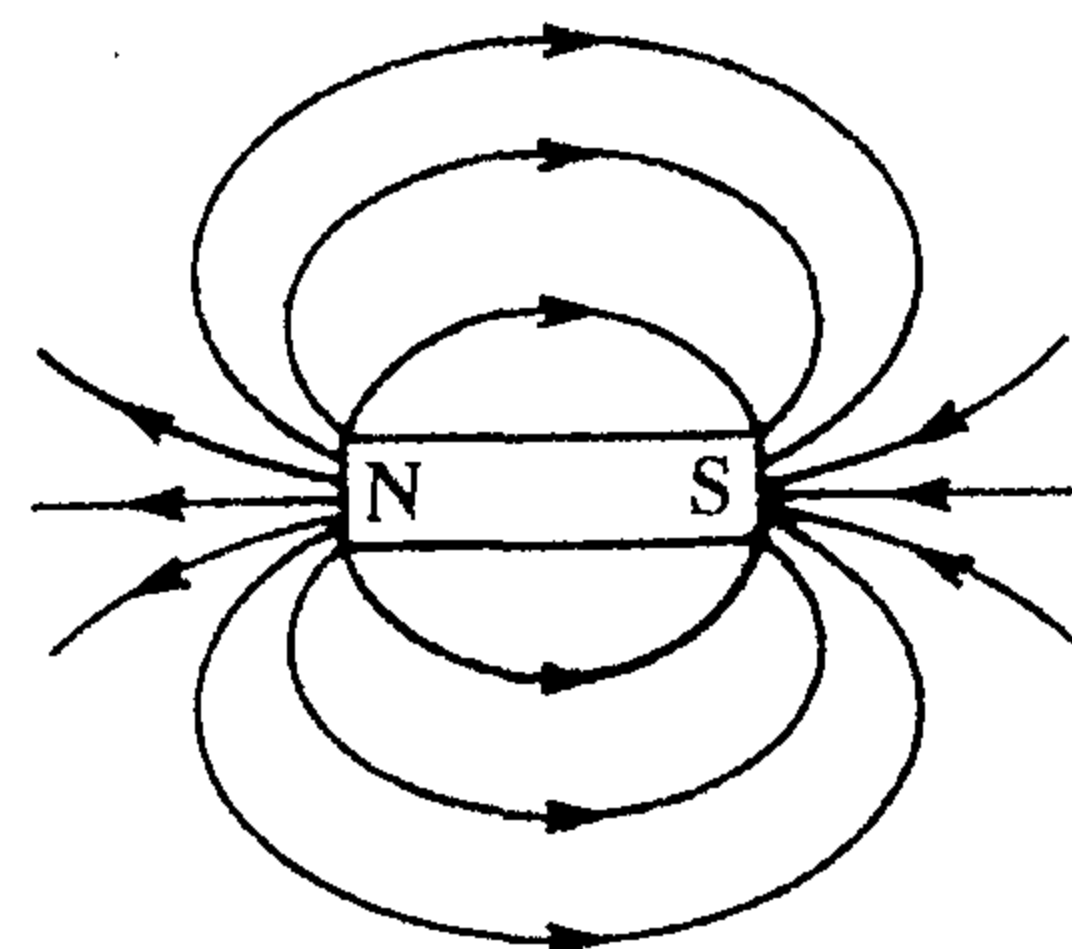


(D)

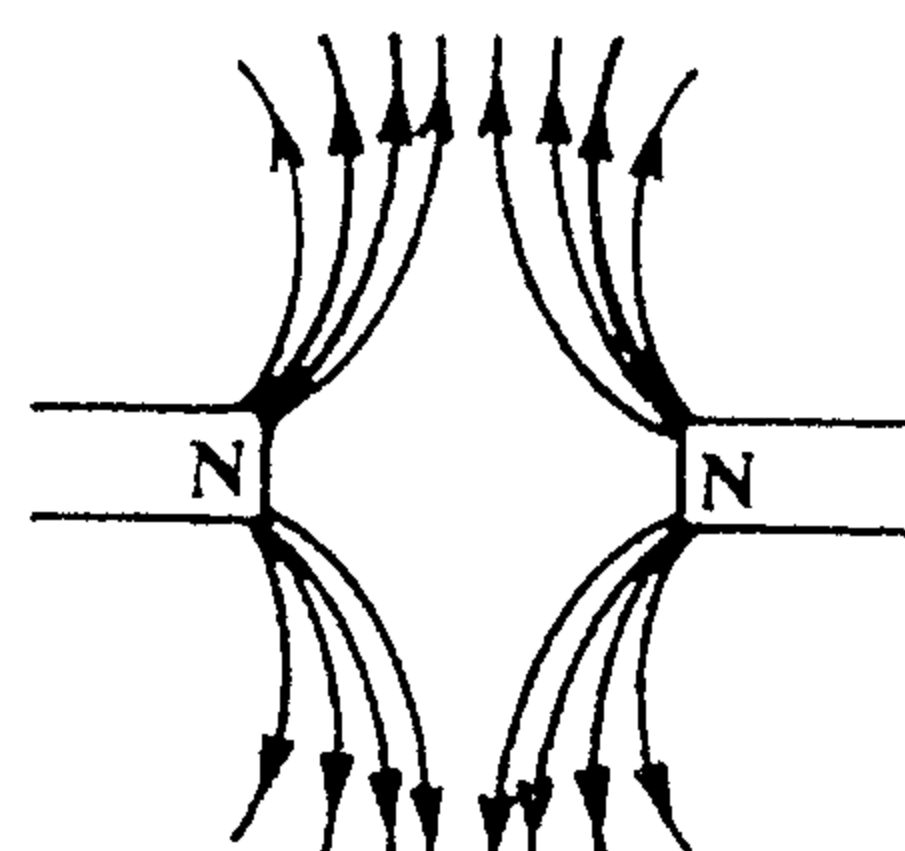


51.

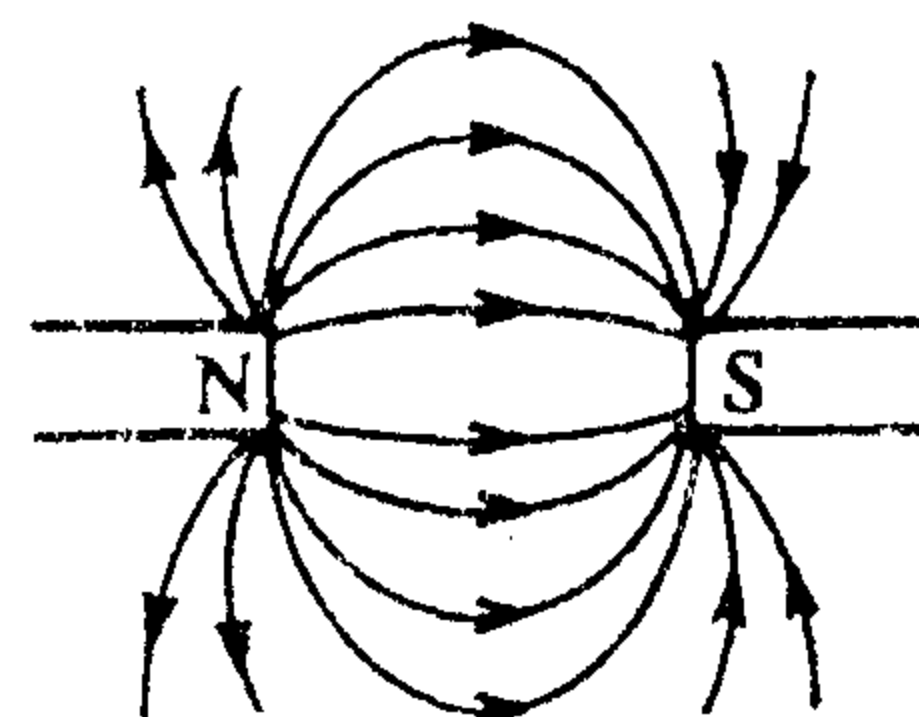
The diagrams below show the magnetic field lines plotted by a student.



I



II



III

Which of the following are correct?

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

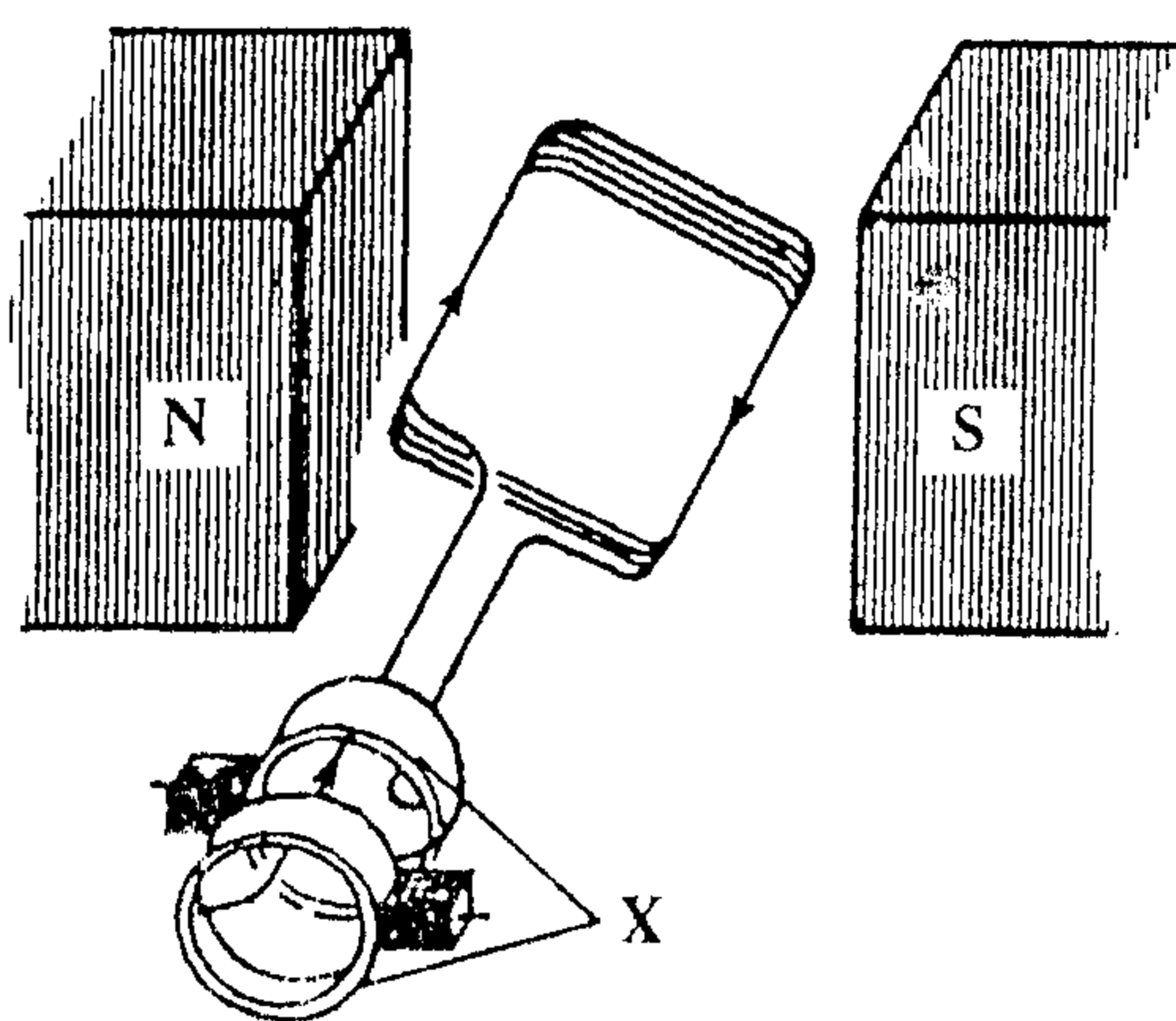
52. Which of the following statements about alternating current is true?

- (A) It can be changed into direct current by a transformer.
- (B) It can be rectified by using a semiconductor diode.
- (C) It can be used to recharge a battery.
- (D) It is used to transmit electrical energy because of its high frequency.

53. A wire carrying a current in a magnetic field may experience a force. Which of the following devices does NOT depend on this force?

- (A) Loudspeaker
- (B) Electric motor
- (C) Moving coil galvanometer
- (D) Electromagnet

Item 54 refers to the following diagram.



54. The parts labelled X in the diagram are known as the

- (A) brushes
- (B) commutator
- (C) armature
- (D) coil

55. The Rutherford model of the atom suggests that an atom consists of a

- (A) solid mass of protons and electrons
- (B) nucleus of protons only with orbiting electrons
- (C) nucleus of equal numbers of neutrons and electrons with orbiting protons
- (D) nucleus of protons and neutrons with orbiting electrons

56. The number of neutrons present in the nucleus of the nuclide $^{222}\text{Rn}^{86}$ is

- (A) 86
- (B) 136
- (C) 222
- (D) 308

57. Which of the following is NOT affected by an electric field?

- (A) Alpha particle
- (B) Beta particle
- (C) Neutron
- (D) Electron

58. Which of the following scientists discovered radium?

- (A) Marie Curie
- (B) Isaac Newton
- (C) Albert Einstein
- (D) J. J. Thompson

59. Which of the following CANNOT be deflected by a magnetic field?

- (A) Alpha particles
- (B) Beta particles
- (C) Gamma rays
- (D) Electrons

60. Which of the following scientists would you associate with nuclear energy?

- (A) Thompson
- (B) Bohr
- (C) Chadwick
- (D) Einstein